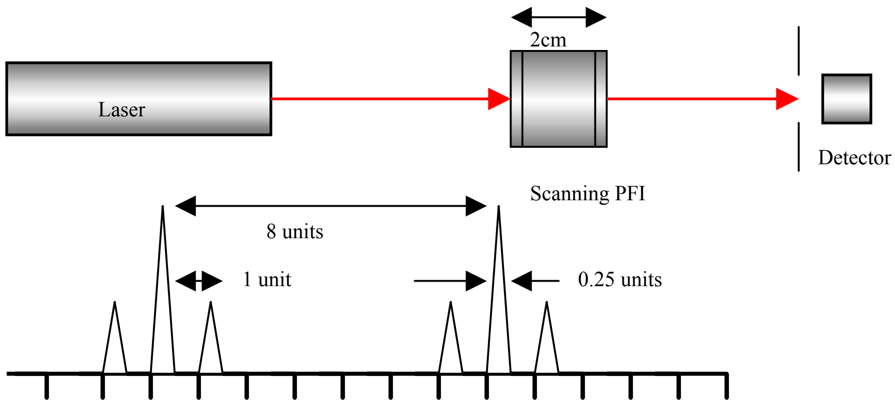
**ECE/PH 482/582, Fall 2003** **Name:**______________________________

**LONGITUDINAL MODES LAB QUIZ– 20 pts** **Last, First**

Shown in the figure below is the experimental setup. A laser is coming from a "black box" into a scanning Fabry-Perot interferometer (FPI). The laser wavelength is 1 micron. All other data are shown on the figure. The scanning F-P output from the oscilloscope is shown also.




1. What is the full scanning length of the Fabry-Perot? How do you know?

2. What is the FSR of the F-P?

3. What is the finesse of the F-P?

4. What is the length of the laser cavity (assume n=1) inside the "black Tube"?

5. If the F-P spacing is expanded to 1 cm, sketch the new F-P output. [Use the “16 units” scale as in the sketch above.]